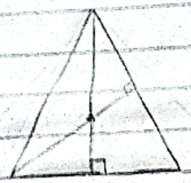
C/E/I:

Triangles

Intro:
Smaller Polygon
$\angle A + \angle B + \angle C = 180°$
$\angle x = \angle A + \angle C$



Centroid



Orthocenter

☆ Similarity of Δ:-

BPT:
If a line parallel to one side of a Δ to intersect the other side is Distinct points, the other 2 sides will divide in same ratio.
For ex-
Given: In ΔABC, DE||BC
DE intersects AB and AC at D and E respectively

To prove: $\frac{AD}{DB} = \frac{AE}{EC}$

Construction: Join BE and CD and EP ⊥ AB and DQ ⊥ AC,

Proof: Since EP ⊥ AB,
∴ EP is the height of ΔADE and ΔDBE.
Thus, $ar(\Delta ADE) = \frac{1}{2} \times AD \times EP$

$ar(\Delta DBE) = \frac{1}{2} \times DB \times EP$

$\therefore \frac{ar(\Delta ADE)}{ar(\Delta DBE)} = \frac{\frac{1}{2} \times AD \times EP}{\frac{1}{2} \times DB \times EP} = \frac{AD}{DB}$ ... ①

Similarly, $\frac{ar(\Delta ADE)}{ar(\Delta DEC)} = \frac{\frac{1}{2} \times AE \times DQ}{\frac{1}{2} \times EC \times DQ} = \frac{AE}{EC}$ ... ②

Since, ΔDBE and ΔDEC are on the same base DE and between the same parallels DE and BC.

$ar(\Delta DBE) = ar(\Delta DEC)$

$\frac{1}{ar(\Delta DBE)} = \frac{1}{ar(\Delta DEC)}$ (Taking reciprocals)

$\frac{ar(\Delta ADE)}{ar(\Delta DBE)} = \frac{ar(\Delta ADE)}{ar(\Delta DEC)}$ (Multiply by $ar(\Delta ADE)$)

Hence $\frac{AD}{DB} = \frac{AE}{EC}$ (From ① & ②)

Converse of BPT:

Given: In ΔABC, $\frac{AD}{DB} = \frac{AE}{EC}$, line l intersects AB and AC at D and E respectively.

To prove: DE||BC

Construction: DF || BC,

Proof: $\frac{AD}{DB} = \frac{AF}{FC}$ (BPT) -①

Since, $\frac{AD}{DB} = \frac{AE}{EC}$ (Given) -②

$\therefore \frac{AF}{FC} = \frac{AE}{EC}$ (From ① & ②)

$\Rightarrow \frac{AF}{FC} + 1 = \frac{AE}{EC} + 1$ (adding 1 on both sides)

$\Rightarrow \frac{AF+FC}{FC} = \frac{AE+EC}{EC}$

$\Rightarrow \frac{AC}{FC} = \frac{AC}{EC}$

$\Rightarrow \frac{1}{FC} = \frac{1}{EC}$

$\Rightarrow FC = EC$ (Dividing by AC)

Since, DF || BC

i.e. || BC or DE || BC

Similarity of Δ:-

Theorem 6.3 (AAA criterion):
If in two triangles, corresponding angles are equal, i.e. the two triangles are equiangular, then the triangles are similar.

In ΔABC and ΔDEF,

$\angle A = \angle D$

$\angle B = \angle E$

$\therefore \Delta ABC \sim \Delta DEF$ (AAA)

Theorem 6.4 (SSS criterion):
If the corresponding sides of two triangles are proportional, then they are similar.

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

$\angle A = \angle D$ ....

$\angle B = \angle E$

$\angle C = \angle F$

Theorem 6.5:
If two sides in one triangle are proportional to two sides in another triangle and the included angles are equal, then the two triangles are similar.

$$\frac{AB}{DE} = \frac{AC}{DF}$$

$\Rightarrow \angle A = \angle D$

Thus, $\angle B = \angle E$

$\Rightarrow \angle C = \angle F$

So, $\Delta ABC \sim \Delta DEF$

Ex-6.1

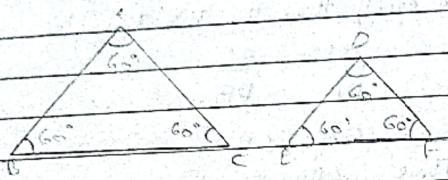
1-i-Ans All circles are similar.

ii-Ans All square are similar.

iii-Ans All equilateral triangles are similar.

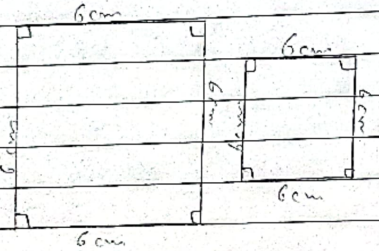
iv-Ans (a) equal (b) proportional

2-i-Ans I



Two equilateral triangles are similar.

II



Two squares are similar.

Ex-6.2

1-Sol-(i) Since, DE || BC

$\Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$ (CBPT)

$\Rightarrow \frac{1.5}{3} = \frac{1}{EC}$

$\Rightarrow EC = 2\text{cm}$

ii-Sol Since, DE || BC

$\Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$ (CBPT)

$\Rightarrow \frac{AD}{7.2} = \frac{1.8}{5.2}$

$\Rightarrow AD = 2.4\text{cm}$

2-i-Sol Here, $\frac{PE}{EQ} = \frac{3.9}{3} = 1.3$ cm

and $\frac{PF}{FR} = \frac{3.6}{2.4} = \frac{3C}{24} = \frac{3}{2} = 1.8$ cm

Thus, $\frac{PE}{EQ} \neq \frac{PF}{FR}$

Hence, EF is not parallel to QR.

ii-Sol Here, $\frac{PE}{EQ} = \frac{4}{4.5} = \frac{40}{45} = \frac{8}{9}$

and $\frac{PF}{RF} = \frac{8}{9}$

Thus, $\frac{PE}{EQ} = \frac{PF}{RF}$

Hence, EF || QR

4-Sol. In $\triangle ABC$, $DE \parallel AC$,

$\therefore \frac{BD}{DA} = \frac{BE}{EC}$ (BPT) $-(1)$

Again in $\triangle ABE$,

$DF \parallel AE$

$\therefore \frac{BD}{DA} = \frac{BF}{FE}$ (BPT) $-(2)$

From (1) & (2)

$\frac{BE}{EC} = \frac{BF}{FE}$

Hence, $\frac{BF}{FE} = \frac{BE}{EC}$

5-Sol. In $\triangle OPQ$, $DE \parallel OQ$,

$\therefore \frac{PE}{EQ} = \frac{PD}{OD}$ $-(1)$

Again in $\triangle OPR$, $DE \parallel OR$

$\frac{PD}{DO} = \frac{PF}{FR}$ $-(2)$

From (1) & (2), $\frac{PE}{EQ} = \frac{PF}{FR}$

Hence, $EF \parallel QR$ (By converse of BPT)

7-Sol. Since, $DE \parallel BC$,

$\therefore \frac{AD}{DB} = \frac{AE}{EC}$ (BPT) $-(1)$

But $AD = DB$

$\Rightarrow \frac{AD}{DB} = 1$

$\therefore (1) \Rightarrow \frac{AE}{EC} = 1$

Hence, $AE = EC$

8-Sol. $AD = DB$

and $AE = EC$

$\Rightarrow \frac{AD}{DB} = 1$

and $\frac{AE}{EC} = 1$

$\Rightarrow \frac{AD}{DB} = \frac{BE}{EC}$

Hence, $DE \parallel BC$ (converse of BPT)

Ex - 6.3

2-Sol (i) $\angle DOC = 180° - 125° = 55°$ (linear pair)

(ii) $\angle DCO = 180° - (70° + 55°)$
$= 180° - 125° = 55°$

(iii) $\angle OAB = \angle DCO = 55°$ (Alt. $\angle$s)

3-Sol In $\triangle OAB$ and $\triangle ODC$,

$\angle 1 = \angle 2$ (Alt. $\angle$s)

and $\angle 2 = \angle 4$ (Alt. $\angle$s)

$\angle AOB = \angle COD$ (Vertically opp. $\angle$s)

$\therefore$ By AA criterion, $\triangle OAB \sim \triangle OCD$

Hence, $\frac{OA}{OC} = \frac{OB}{OD}$

4-Sol In $\triangle ABC$ and $\triangle AMP$,

$\angle A = \angle A$ (Common)

$\angle B = \angle M$ (Each 90°)

$\triangle ABC \sim \triangle AMP$ (AA)

(i) Since, $\triangle ABC \sim \triangle AMP$,

$\therefore \frac{AC}{AP} = \frac{BC}{MP}$

Hence, $\frac{CA}{PA} = \frac{BC}{MP}$

11-Sol Since ABC is an isosceles $\triangle$,

$\therefore AB = AC$

and $\angle B = \angle C$

In $\triangle ABD$ and $\triangle ECF$,

$\angle ABC = \angle ECF$ (Proved above)

$\angle ADB = \angle EFC$ (Each 90°)

Hence, $\triangle ABD \sim \triangle ECF$ (AA)

12-Sol In $\triangle ABC$ and $\triangle ADC$,

$\angle BAC = \angle ADC$ (Given)

$\angle ACD = \angle ACD$ (common)

$\therefore \triangle ABC \sim \triangle DAC$ (AA)

$\therefore \frac{BC}{AC} = \frac{AB}{AD} = \frac{AC}{DC}$

$\Rightarrow \frac{BC}{AC} = \frac{AC}{DC}$

Hence, $AC^2 = BC \times DC$

$\Rightarrow CA^2 = CB \times CD$

15-Sol Let AB be the vertical pole and AC be its shadow.
Let PQ be the vertical tower and PR be its shadow.

Let $PQ = x$ cm

Thus, $AB = 6$ cm,

$AC = 4$ m

and $PR = 28$ m

Now, $\angle C = \angle R$ (Each elevation of sun is same)

$\angle A = \angle P$ (Each 90°)

$\triangle ABC \sim \triangle PQR$ (AA)

$\therefore \frac{AB}{PQ} = \frac{AC}{PR}$

$\frac{6}{x} = \frac{4}{28}$

$x = 42$

Hence, the height of the tower is 42 m.

16-Sol Here, $\triangle ABC \sim \triangle PQR$

$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{2BD}{2QM} = \frac{BD}{QM}$ (D is the mid-point of BC and M is the mid-point of QR)

Also, $\angle B = \angle Q$

In $\triangle s$ ABD and PQM,

$$\frac{AB}{PQ} = \frac{BD}{QM}$$

and $\angle B = \angle Q$

$\therefore \triangle ABD \sim \triangle PQM$

Hence, $\frac{AB}{PQ} = \frac{AD}{PM}$